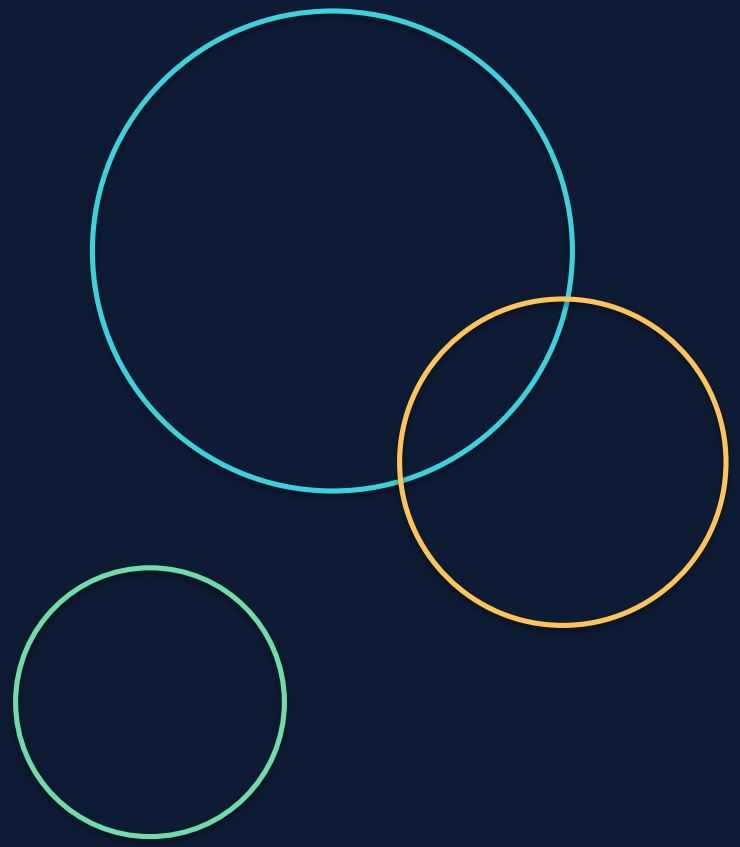
TECHNICAL PRESENTATION

CRYPTOGRAPHY: CIPHERS, THREATS & MATHEMATICS



From Caesar and Vigenère to steganography, security goals, and number theory

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1. From Caesar to Vigenère

Evolution of substitution ciphers

Caesar Cipher

Each letter shifts by a fixed number.

Example: HELLO → KHOOR (shift +3).

Simple, but easy to break.

Monoalphabetic Cipher

Each plaintext letter maps to one fixed substitute.

More possible mappings, but letter-frequency patterns remain.

Vigenère Cipher

Uses a repeating keyword to apply different shifts.

Example key: KEY.

Harder than Caesar, but vulnerable to modern analysis.

Key takeaway

Substitution replaces symbols. Vigenère improves on a single fixed shift by changing the shift according to a key, but these classical ciphers are not secure for modern use.

2. Substitution vs Transposition

Two classic ways to transform a message

Substitution

- Changes the identity of characters
- Example: CAT → DBU with Caesar shift +1
- Letter order stays the same
- Security depends on the cipher and key

Transposition

- Rearranges character positions
- Example: MEET → EMET (illustrative rearrangement)
- Characters stay the same, order changes
- Patterns may still leak without strong design

Which is more secure? Neither is inherently better. Classical versions of both are weak alone; modern systems combine strong algorithms, keys, and careful implementation.

3. Attacks, Threats, Risks & Security Goals

Understand what can go wrong—and what security protects

Attack

An action that attempts to harm or access a system.

Example: phishing email.

Threat

A possible cause of harm.

Example: an attacker stealing login details.

Vulnerability

A weakness that can be exploited.

Example: reused password.

Risk

Likelihood × impact of harm.

Example: account takeover and data loss.

The CIA security goals

Confidentiality

Only authorized people can see data.

Integrity

Data remains accurate and unaltered.

Availability

Systems and data are accessible when needed.

4. Steganography vs Cryptography

Hiding a message versus hiding its meaning

Cryptography

Goal: make message contents unreadable.

Plaintext → encryption + key → ciphertext.

Example: HELLO → KHOOR (Caesar illustration).

Encrypted data may be noticed.

Steganography

Goal: conceal that a message exists.

Secret data is embedded in an image, audio, or video.

Example: text hidden inside a photo.

The carrier may look ordinary.

Layered protection: encrypt the message first, then hide the ciphertext inside a carrier file. This can protect both the content and the presence of the message, but does not guarantee security.

5. Number Theory: From Euclid to Euler

Mathematics behind important cryptographic methods

Prime Numbers

A prime has exactly two positive divisors: 1 and itself.

Example: 2, 3, 5, 7, 11.

Large primes are important in RSA key generation.

Euclidean Algorithm

Finds the greatest common divisor (GCD).

Example: gcd(18, 12) = 6.

Used in calculations related to modular inverses.

Euler's Totient & Theorem

φ(n) counts positive integers up to n that are coprime to n.

Euler's theorem: $a^{\varphi(n)} \equiv 1 \pmod{n}$, when $\gcd(a,n)=1$.

Why it matters: modular arithmetic, prime factorization, GCDs, and modular inverses help build and explain public-key cryptography such as RSA. Real systems use carefully selected parameters and proven implementations.

Putting the Concepts Together

A simple secure-message scenario

1	**Message**	A student wants to send private information.
2	**Encrypt**	Cryptography converts readable text into ciphertext.
3	**Hide (optional)**	Steganography embeds ciphertext in a photo.
4	**Transmit**	The file is sent over a network.
5	**Recover**	The receiver extracts and decrypts the message.

Note: use established encryption; classical ciphers are for learning, not real sensitive data.

Choosing the Right Protection

What each concept contributes

Technique	Protects	Example / limitation
Substitution	Symbol identity	Caesar / Vigenère; classical methods are breakable
Transposition	Symbol order	Rearranges letters; patterns can remain
Cryptography	Message meaning	Modern encryption protects confidentiality
Steganography	Message visibility	Hidden data may be detected or damaged
Number theory	Mathematical foundations	Supports RSA and modular calculations

Security is not one technique: it depends on the threat, the data, the keys, and the correct implementation.

Mini Case Study: Protecting a College Portal

Map risks to security goals and controls

Scenario

A college portal stores student marks and personal details.

Threat: an attacker tries to access accounts.

Risk: data exposure or altered marks.

Security Goals

Confidentiality: restrict access to authorized users.

Integrity: prevent unauthorized changes to marks.

Availability: keep the portal working during exams.

Practical Controls

Use strong unique passwords and multi-factor authentication.

Encrypt data in transit and at rest.

Keep backups, updates, access logs, and recovery plans.

Lesson: identify threats and weaknesses, estimate the risk, then select controls that support confidentiality, integrity, and availability.

Conclusion & Key Takeaways

Five topics, one goal: protect information

- Caesar and Vigenère show how substitution ciphers evolved.
- Substitution changes symbols; transposition changes their order.
- Attacks exploit weaknesses and can threaten confidentiality, integrity, and availability.
- Cryptography protects message content; steganography hides the presence of a message.
- Number theory provides mathematical foundations for public-key cryptography.

Final Message

Use modern, well-reviewed cryptographic tools for real security. Classical ciphers are useful for learning, not for protecting sensitive data.

References for further study

- William Stallings, Cryptography and Network Security
- NIST Cryptographic Standards and Guidelines
- Computer Security: Principles and Practice

THANK YOU •